

Feature handling & observed behavior (+ folder provided)
- for both full features and borrower-side

- **Baseline (outputs/)**
 - Features: uses all engineered columns except IDs are ignored by the pipeline. History signals remain.
 - Behavior: Very high AUC (~0.99) driven partly by strong history features; likely overstates real-world performance if the same borrowers appear in train/test.
 - Metrics logged: accuracy, precision, recall, F1, AUC-ROC, AUC-PR.
 - Observed: **full** best AUC 0.995 (XGBoost); best F1 0.827 with LightGBM (precision 0.810, recall 0.846, accuracy ~0.994). **borrower** best AUC 0.970; best F1 0.565 with LightGBM (precision 0.466, recall 0.721, accuracy ~0.980).
- **Drop history/leakage (outputs_drop_features/)**
 - Features dropped: borrower history and repayment-derived signals such as `num_previous_loans`, `num_previous_defaults`, `past_default_rate`, `trend_in_amount/burden`, `repayment_intensity/consistency`, `burden_percentile`, IDs.
 - Behavior: Metrics dip vs. baseline but stay strong; LightGBM still leads. More realistic when past-behavior is weak or unavailable.
 - Metrics logged: accuracy, precision, recall, F1, AUC-ROC, AUC-PR.
 - Observed: **full** CatBoost AUC 0.995; LightGBM F1 0.847 (precision 0.816, recall 0.881, accuracy ~0.994). **borrower** LightGBM F1 0.578 (precision 0.476, recall 0.736, accuracy ~0.980; AUC 0.971).
- **First-time borrower emphasis (outputs_drop_features_first_time_borrower/)**
 - Features dropped: same as above, focusing on removing past-behavior clues to mimic thin-file borrowers.
 - Behavior: Slightly lower lift than baseline; LightGBM remains the most reliable. Useful for onboarding new borrowers with little history.
 - Metrics logged: accuracy, precision, recall, F1, AUC-ROC, AUC-PR.
 - Observed: **full** CatBoost AUC 0.996; LightGBM F1 0.824 (precision 0.781, recall 0.871, accuracy ~0.993). **borrower** LightGBM F1 0.585 (precision 0.467, recall 0.781, accuracy ~0.980; AUC 0.963).
- **High recall (outputs_high_recall/)**
 - Features: full set; threshold is tuned to hit recall ~90%.
 - Behavior: Catches most defaulters, but precision falls (more good borrowers flagged). Choose when “don’t miss risk” outweighs extra reviews/declines.
 - Metrics logged: accuracy, precision, recall, F1, AUC-ROC, AUC-PR, chosen threshold.
 - Observed: **full** LightGBM F1 0.825 at recall 0.900, precision 0.761 (accuracy ~0.993; AUC 0.994). **borrower** LightGBM F1 0.355 at recall 0.900, precision 0.221 (accuracy ~0.940; AUC 0.970).
- **Leakage-safe (outputs_leakage_safe/)**
 - Features dropped: repayment-derived/leaky signals (e.g.,

- `repayment_intensity`, `repayment_consistency`, `latest_amount_ma3`, `trend_in_amount/burden`, `burden_percentile`) plus IDs.
- Split: group by `customer_id` (or time-based if a date column exists) so no borrower is in both train and test.
- Behavior: AUC/F1 remain strong but slightly lower than baseline—closer to true production performance. CatBoost edges out others, LightGBM/XGBoost close.
- Metrics logged: accuracy, precision, recall, F1, AUC-ROC, AUC-PR, chosen threshold.
- Observed: **full** CatBoost AUC 0.994, F1 0.841 (precision 0.857, recall 0.825, accuracy ~0.994). **borrower** CatBoost AUC 0.950, F1 0.350 (precision 0.224, recall 0.801, accuracy ~0.940); LightGBM F1 0.312 as next best.

Metrics reported (used in every folder)

- **accuracy**: share of all loans the model labeled correctly (defaults and non-defaults).
- **precision**: of the loans flagged as risky, the share that truly default.
- **recall**: of the loans that defaulted, the share the model caught.
- **f1**: balance of precision and recall (higher = better balance).
- **auc_roc**: how well the model separates good vs. bad loans across all score cutoffs.
- **auc_pr**: similar separation metric but focused on the positive (default) class in imbalanced data.
- **threshold**: only in high-recall/leakage-safe runs; the score cutoff chosen to balance business goals (higher recall vs. fewer false alarms).

Takeaways

- Full feature set: Gradient boosting models (LightGBM/CatBoost/XGBoost) separate good vs. bad loans very well (AUC ~0.99). LightGBM tends to give the best balance of precision vs. recall; CatBoost is close behind.
- Borrower-only feature set: Performance drops (as expected) but LightGBM still leads. Expect more false alarms or missed defaults when history is limited.
- High-recall setup: Successfully drives recall toward ~90%, useful when the business priority is “don’t miss bad loans,” at the expense of more good borrowers being flagged.
- Leakage-safe split: Results are slightly more conservative than the baseline, reflecting a fairer test. CatBoost edges out others, with LightGBM and XGBoost close.
- First-time borrower scenarios: Stripping history signals reduces lift, but LightGBM remains the most reliable generalist across tests.

How to read the outputs

- Each folder has a `metrics_summary.csv` with the key numbers:
 - **precision**: of loans we flag, how many truly default.
 - **recall**: of all defaults, how many we catch.
 - **auc_roc/auc_pr**: overall ranking quality (higher is better).
 - **threshold** (high-recall/leakage-safe runs): the cutoff used to trade off risk vs. approvals.
- PNGs (`roc_*.png`, `pr_*.png`, `confusion_*.png`) visualize trade-offs; `classification_report_*.txt` shows per-class precision/recall.